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Changes in nitrogen loads to estuaries following implementation of governmental action plans in Denmark: A paired catchment and estuary approach for analysing regional responses

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ABSTRACT

Since the late 1980s a wide range of regulatory measures, addressing both point sources and diffuse N pollution from agricultural land, have been implemented in estuary catchments to reduce the land based nitrogen (N) load of the Danish aquatic environment. Monitoring has been undertaken of key agro-environmental N indicators for different parts of the DPSIR concept (Driving forces, Pressures, State, Impact and Responses for N) in 10 catchments (covering 35% of the Danish land area) and their estuaries during the period 1990–2009. The N load to these 10 estuaries has been reduced by 39% in total, mostly (28%) by reducing the N loss from diffuse sources. For all the 10 linked catchments and estuaries decreasing trends have been reported for various indicators, amounting to 40–53% for the catchment N surplus, *ca.* 30% for the efficiency of use of cattle slurry N and *ca.* 40% for pig slurry N, 18–55% for the mean flow-weighted total N concentrations in inlet water to estuaries, and 24–62% for total N concentrations in the upper and middle reaches of the estuaries. In eight catchments and their estuaries we found a direct response (<5 years) of the diffuse N loads from catchments to reductions in the agricultural N surplus, the response differing between catchments because of wide variations in N removal in groundwater and surface waters. In only two of the catchments and estuaries a time delay of N in groundwater aquifers was a major factor (decadal), possibly due to oxic groundwater in chalk aquifers. Knowledge of responses to agricultural N regulation is highly valuable for decision makers and catchment managers working with the implementation of EU directives such as the Water Framework Directive. Based on our findings we suggest that targeted and catchment-specific measures are the most cost-efficient way to achieve good chemical quality in estuaries.

1. Introduction

Use of nitrogen (N) in agricultural crop production in the form of fertiliser and animal manure is increasing world-wide (OECD, 2008). Future demands for increases in the agricultural production of crops and animal products for a growing world population will most likely augment the use of N. The increases in N consumption in European agriculture during the 1940s to the 1980s had serious deleterious effects on the water quality in both groundwater and surface water bodies (Kronvang et al., 1993) and came to constitute an increasing

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threat to European drinking water supplies (EEA, 2007). In response, the European Community adopted the Nitrates Directive in 1991 to assist in the protection of drinking water quality (EC, 1991), and the use of N in fertilisers in European countries has thus decreased since 1990 (Bouraoui and Grizzetti, 2011).

The N threat to European drinking water supplies was not the sole problem, as the N losses to surface, particularly marine, waters led to severe eutrophication problems in the form of excess algal blooms, loss of benthic vegetation and spread of hypoxia (Kronvang et al., 1993, 2005; Conley et al., 2000; Artioli et al., 2008). The eutrophication problems in the Baltic Sea were addressed by the Helsinki Commission (HELCOM), and plans to monitor, manage and reduce nutrient loads to the Baltic Sea by 50% before 1995 were adopted in 1988 by the countries around the Baltic Sea (HELCOM, 1988). HELCOM revised the 1988 declaration in 1992 with the aim to accelerate the implementation of measures to substantially reduce nutrient loading. Following the revision, national plans have been adopted to combat the nutrient pollution; an example being Denmark, which during the last 25 years has implemented several Nutrient Action Plans (NAPs) (Kronvang et al., 2008).

The aim of this paper is to show the outcome of using the DPSIR concept (Driving forces, Pressures, State, Impact and Responses) linked to key agro-environmental indicators monitored for each DPSIR level to identify the linkages between climate, political decisions regarding NAPs, costs and the observed trends in indicators at the different DPSIR levels (Fig. 1).

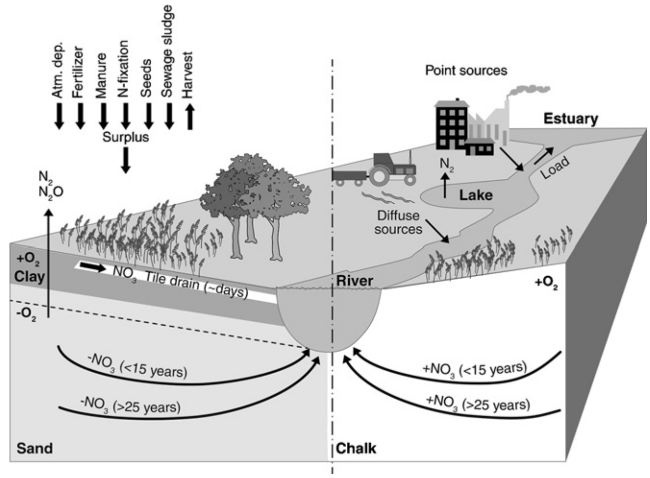

Fig. 1 – Conceptual diagram showing nitrogen input and surplus on agricultural land, nitrogen sources, attenuation and delay in two different groundwater aquifers in Denmark.

2. History of NAPs in Denmark

Denmark has adopted six nitrogen action plans (NAPs) since the mid-1980s. The first NAP was the NPO-action plan initiated in 1985 as a response to groundwater monitoring results showing increasing concentrations of nitrate in groundwater aquifers (DEPA, 1983). The second NAP was adopted by the Danish Parliament in 1987 (NAPI) after fishermen landed dead lobsters from the Kattegat (at the entrance to the Baltic Sea), which was widely broadcasted in national television. NAPI was the first action plan to stipulate that the national load target for N (50% reduction) should be achieved within a specific time period (*ca.* 5 years) (Kronvang et al., 1993). The national aquatic monitoring programme established together with NAPI did not, however, reveal any clear declining trends in N leaching and loads to surface water bodies after the 5-year period, and a new action plan on sustainable agriculture (APSA) was therefore adopted by the Danish Parliament in 1991 (Kronvang et al., 2008). One of the major reasons why the monitoring programme did not trace significant changes in N leaching and loads to the aquatic environment was large inter-annual variations in N loads, and possibly also to a lesser degree attenuation in groundwater (Fig. 1). NAPII was implemented by the Danish Parliament in 1998, and in 2003 NAPIII was adopted and included for the first time mitigation measures against diffuse phosphorus losses (Kronvang et al., 2008). Most recently, in 2012, the WFD action plan was adopted to enforce some of the measures needed to fulfil the objectives of the EU Water Framework Directive (WFD). This plan includes, for the first time, catchment specific load targets to each estuary and a suite of targeted mitigation options (wetland restoration, buffer strips, catch crops). Since 1990, the total land-based N load to Danish coastal waters has been reduced by *ca.* 50% (Kronvang et al., 2008). On a national scale, the N discharges from point sources have been reduced by 18,500 t N (77%) during the period 1990–2009, and the diffuse N load has decreased from 81,000 t N yr^{-1} to 49,000 t N yr^{-1} (40%), the latter due to the adopted general regulations of Danish agriculture reducing the N surplus (Table 1).

Table 1 – National figures on changes in nitrogen created by reductions of the nitrogen discharges from point sources to surface waters, the nitrogen surplus in Danish agriculture and increased N retention in surface waters due re-established wetlands under the Danish Nitrogen Action Plans (Jacobsen et al., 2004).

National figures	N load, 10^3 t N/y		Reduction, 10^3 t N/y	Costs	Euro/kg N reduced
	1990	2009	1990–2009	billion €	
Point sources	24	5.5	18.5	0.97	52
General measures					
Nitrogen surplus (fields)	412	205	207	2.571	13
Targeted measures					
Re-establishment of wetlands and lakes			1.82	0.09	49

Table 2 – N surplus reductions of 2,10,000 t N in fertilisers and 19,000 t N in manure specified for general and target measures and land use changes implemented within the framework of five national action plans.

Measure	NAP 1989–2009	Reduction of N surplus (1000 tonnes N)	
		Fertilisers (%)	Manure (%)
General measures			
Regulation of nitrogen fertiliser and manure			
Manure storage, 6–9 months	NAPI	50	
Crop fertilisation plans	NAPI		
Increased utilisation of manure	NAPSA[a]		
	NAPII[b]		
	NAPIII[c]		
Utilisation of livestock fodder	NAPII	−7	100
Change in livestock: more pigs/less dairy cows	NAPIII		
N fertiliser, 90% of economic optimum[b]	NAPII	19	
Catch crops[b]	NAP II	2	
Change in land use			
Afforestation	NAPII	0	
	NAP III	0	
Set aside and decrease in arable land	NAPSA[a]	21	
Protection of groundwater wells[b]	NAPII	5	
Conversion to organic dairy production[b]	NAPII, III	8	
Target measure			
Re-establishment of wetlands	NAPII, III	2	

[a] Ministry of Agriculture (1991).
[b] Iversen et al. (1998).
[c] Waagepetersen et al. (2009).

The general and targeted mitigation measures adopted in Denmark according to the different action plans and their estimated effect on fertiliser use and manure consumption are shown in Table 2. A more efficient use of nitrogen in manure reduced the fertiliser consumption by 50%, and EU-regulated set aside areas and structural changes in the size of arable land decreased the consumption by 21%. A 10% cut in the optimal N quota to crops (N quota is the specific N standard to each grown crop) as defined from the economic optimum before NAPII in 1998 resulted in a 19% decline in the consumption of fertilisers (Table 2). Minor consumption reductions were related to ceased application of fertilisers in protection areas around groundwater wells, planting of catch crops, conversion to organic dairy production and re-establishment of wetlands. Afforestation does not decrease fertiliser consumption as the national consumption is regulated by the N quota to crops. More efficient utilisation of livestock fodder and changes in livestock composition to more pigs and less dairy cows led to a reduction in manure production of about 19,000 t N during the period 1989–2009 (Iversen et al., 1998).

Implementation of the actions plans has been and is still costly for industry, farmers and individual citizens. Some of the investments in measures – especially manure storage facilities – are amortised over perhaps a 10-year period, rendering it irrelevant to calculate the investment in cost per year, and the costs of the five action plans have therefore been estimated as total costs. Information on the total expected investments for the different action plans has been sourced from the existing literature (Jacobsen et al., 2004). Thus, for NAPI, the expected investment in treatment of point source pollution from public sewage and industrial emissions amounted to 0.97 billion Euro, a total investment in agricultural measures of 2.71 billion Euro covered mainly the costs of establishing manure and slurry storage facilities and the lower crop production resulting from the EU-stipulated set aside scheme (Jacobsen et al., 2004). Also general measures such as catch crops and afforestation as well as targeted measures such as re-establishment of wetlands were implemented in connection with the introduction of NAPII and -III, but compared to the total effect and costs of all action plans they have proved to be less significant for N load reductions to estuaries (Jacobsen et al., 2004).

3. Materials and methods

3.1. *Characteristics of study catchments and estuaries*

Ten catchments and their receiving estuaries covering different parts of Denmark were included in this study (Fig. 2). The catchment areas range between 184 and 3477 km^2 and the surface area of the estuaries ranges between 26 and 316 km^2 (Table 3). In total, the 10 catchments cover 15,200 km^2 or 35% of the total Danish land area. The catchments are dominated by agricultural land and the soil type varies from coarse sandy soils in the western catchments to more loamy soils in the east. Freshwater runoff is highest from catchments in the western part of Denmark and gradually decreases towards the east due to a decreasing gradient in rainfall (Table 3).

3.2. *Calculation of nitrogen surplus and use of nitrogen in manure*

The soil surface N balance is used as an environmental indicator (Parris, 1998). Basically, the method assesses the

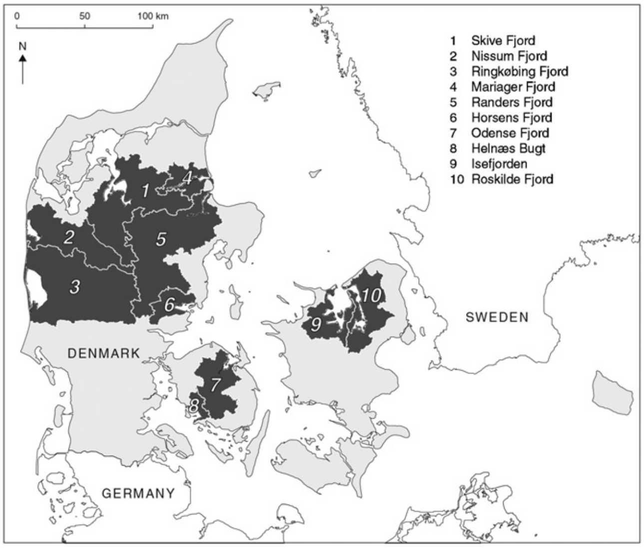

Fig. 2 – Map of Denmark showing the 10 studied catchments and estuaries.

difference between the annual quantity of N entering the soil and the annual quantity of N leaving the soil. The gross N balance includes all emissions of N compounds into soil, water and air and is in this study given as the N surplus calculated for 6 regions of Denmark. Information on the input of N from commercial fertilisers is based on yearly statistics evidencing the amount of fertilisers sold by Danish importers and producers to Danish farmers (Statistics Denmark, 1990–2010). Data on manure is derived from the national assessment of excretion from different animal categories multiplied by the nation-wide number of animals corrected for the ammonia emission from manure in storage tanks and stables (Statistics Denmark, 1990–2010). Estimations of N fixation in legumes are based on the principles of Kyllingsbæk and Hansen (2007) whose model uses dry matter yield as input as well as parameters for N concentrations in dry matter, and the proportion of the N in the legume that is derived from the atmosphere. Input of N from atmospheric deposition is calculated by multiplying the area covered by forest and natural areas by a modelled (10 km × 10 km grid) mean N deposition per hectare. Annual net deposition is measured and modelled in The National Air Monitoring Program (Ellerman et al., 2011). The N deposition on arable soils is calculated from the deposition on natural areas but to added the deposition arising from the extra ammonia emission from manure storage and spreading.

Other sources of N to agricultural soils are seeds and sewage sludge. The N input from seeds is about 2 kg N ha^{-1} as estimated from farm gate balances. Calculations of total N in sewage sludge and waste from industrial products used on agricultural soils are based on data from The Danish Environmental Protection Agency (Mikkelsen, 2003). The N offtake by harvested products was calculated by multiplying the total national assessed yield for different crops (Statistics Denmark, 1990–2010) by the N content calculated from the protein concentration in the crops taken from the Danish tables of feeding recommendations (Kyllingsbæk and Hansen, 2007).

The efficiency of N use in manure is calculated for cattle and pig slurry from farming enterprises located in 5 agro-ecosystems included in the NOVANA programme (national monitoring and assessment programme for the aquatic environment) (Grant et al., 2011). The use of N fertiliser and the required efficiency of manure-N use must not exceed the farm N quota, the efficiency of manure-N use being calculated as:

$$\text{Efficiency of manure-N use}\,(\%) = \left(\frac{\text{Farm N quota to crops} - \text{N in mineral fertiliser}}{\text{N in animal manure}}\right) \times 100\% \qquad (1)$$

3.3. *Calculation of stream nitrogen load*

Following the adoption of NAPI in 1987, a nation-wide monitoring programme for the aquatic environment was established and initiated in 1989 and more than 20 years of monitoring data on all major Danish water bodies are presently available (Kronvang et al., 1993). In this study we use data collected in the investigated catchments within the framework of the stream monitoring programme where discharge and nutrient concentrations are measured at a number of monitoring stations in each catchment (Table 2). Discharges and N loads from between 22% and 87% of the studied catchment areas are gauged (Table 2). Instantaneous discharge (Q) is measured 12–20 times per year using a low friction propeller, and daily discharge values are calculated using relationships between Q and continuously measured fluctuations in stream water levels (*H*). Water sampling at the

Table 3 – Description of the 10 investigated catchments and estuaries.

Estuary	Mean freshwater runoff 1990–2009 (mm yr^{-1})	Catchment area (km^2)	Gauged area (%)	Agricultural area (%)	Sandy soils (%)	Estuarine area (km^2)
1. Skive Fjord	321	2621	67	68	90	248
2. Nissum Fjord	451	1615	68	63	86	63
3. Ringkøbing Fjord	453	3477	67	63	92	281
4. Mariager Fjord	240	572	39	66	91	46
5. Randers Fjord	349	3255	87	62	70	26
6. Horsens Fjord	327	520	56	69	40	80
7. Odense Fjord	282	1060	67	63	48	61
8. Helnæs Bugt	266	184	43	67	63	67
9. Isefjord	194	770	22	68	45	316
10. Roskilde Fjord	174	1182	65	52	44	124

monitoring stations is usually conducted every fortnight and analysed for total N concentrations according to Danish standards. The nitrogen load in the gauged streams is calculated using a linear interpolation method for nutrient concentrations and multiplying the results by average daily discharges (Kronvang and Bruhn, 1996).

Land-based annual N loads and freshwater discharge to the estuaries for the period 1990–2009 were estimated using data from the gauging stations in the catchments combined with modelled freshwater discharge and total N loads for the ungauged part of the catchment according to Windolf et al. (2011). Total N discharges from point sources in the entire catchment area were based on measured and calculated data obtained from the national aquatic and terrestrial monitoring programmes (Andersen et al., 2006).

3.4. *Nitrogen concentrations in estuaries*

The national monitoring programme for estuaries, coastal and open marine waters was initiated in 1989 and the present study includes data from 1989–2009. Total N concentrations were measured from discrete water samples taken from the surface layer, as well as from the bottom layer if the water column was stratified. For most of the estuaries monitoring stations were located to represent upper, middle and lower reaches. Sampling frequencies varied from 12 to 46 times per year. Mean annual total N concentrations were calculated for each station using a general linear model with seasonal and trend components (see Carstensen et al., 2006 for more details).

3.5. *Statistical calculations*

Annual flow-weighted concentrations of N surplus ($N_{surplus}$) were calculated based on the annual catchment field surplus (kg) divided by the annual freshwater discharge to the estuary (10^3 m^3). Annual flow-weighted concentrations of total N in inlet waters to estuaries (N_{inlet}) were calculated based on the sum of monitored and modeled annual total N loads (diffuse and point sources) divided by the annual freshwater runoff from the catchments. The fraction of these concentrations deriving from point sources was estimated as total point source discharges divided by annual freshwater runoff (N_{point}). Trend analysis of data was performed as a linear regression analysis using SAS software version 9.3. Normalisation of the area specific diffuse N load (kg N ha^{-1} yr^{-1}) from the catchment to each estuary was undertaken to account for large inter-annual variations in precipitation and discharge and to facilitate comparisons between catchments. Normalisation was conducted by constructing a catchment-specific linear runoff/load (QT) relationship for log-transformed water runoff (Q for the agrohydrological year (mm yr^{-1})) and the log-transformed calculated area-specific total N load from diffuse sources (*T*, (kg N ha^{-1} yr^{-1})) for the same year. The relationship for data from the period 1990–1997 is thus estimated as:

$$\ln T = a + \beta \ln(Q) \quad (2)$$

after which the logarithm-transformed and runoff normalised relation can be expressed as:

$$\ln T_{norm} = \ln T - \beta \cdot (\ln Q - \ln(Q_{mean})) \quad (3)$$

where T (kg N ha^{-1}) is the total N area specific load from diffuse sources in a given year, Q is water runoff (mm) in the same given year, and β is the estimated slope. Q_{mean} in the equation is average annual runoff (mm) for the whole period used for normalisation.

Subsequently, back transformation is performed (Ferguson, 1986) to calculate the runoff-normalised area specific annual N load (T_{norm}) from diffuse sources (kg N ha^{-1} yr^{-1}):

$$T_{norm} = K_{kor} \exp(\ln T_{norm}) \quad (4)$$

where K_{kor} corrects for the use of log-transformed data to estimate T_{norm}. K_{kor} varies for the individual catchments for which runoff-normalised N load is determined. K_{kor} is calculated as:

$$K_{kor} = \exp(\textbf{half;} \times \text{MeanSquareError}) \quad (5)$$

Ideally, the impact of the year-to-year fluctuations in water runoff is not reflected in the runoff-normalised diffuse N load. However, the method is not well adapted for generating a completely realistic value for very dry years such as 1995/96. If point source discharges (urban waste water treatment plants, industrial plants, fish farms and urban storm waters) from the individual years are added to the runoff-normalised diffuse N load (agriculture and background load), an estimate is obtained for the total N input for the given year at average annual water runoff.

4. Results and discussion

4.1. *Trends in key agro-environmental indicators*

The indicator for the efficiency of manure-N use reflects the history of NAPs in Denmark (supplementary material B). The N efficiency of cattle slurry improved from *ca*. 45% in 1995 to more than 75% in 2009, which is slightly higher than the statutory utilisation norms (norms for use of N in manure within the farm fertiliser budgets required by law) for cattle slurry in 2009. Similarly, the N efficiency of pig slurry increased from *ca*. 40% to 80% during the period 1995–2009, again a slightly higher utilisation than that of the statutory norms (supplementary material B). The improved N use efficiency was a direct effect of the NAPs in which restrictions were made to the effective use of nitrogen in fertiliser and manure stated in the N quota to crops. The efficiency of manure-N use decreased somewhat after 1998 as farm N quota was reduced to 90% of the economic optimum in NAPII (Grant et al., 2011).

Efficient use of N in manure is facilitated by higher storage capacity of slurry (9 months), changed timing of spreading (spring application and a ban on slurry spreading between harvest and 1st February since 1991) and improved spreading equipment reducing ammonia volatisation.

N surplus and stream N load from both point and diffuse sources determined as average flow-weighted concentrations for the 10 catchments were estimated as means for the 3 years in the beginning (1990–1992) and at the end of the study period (2007–2009) (Fig. 3A) Significant negative trends (1990–2009) were found for the catchment N surplus in all 10 catchments (40–53%; supplementary material C). Similarly, a significant negative trend in the catchment N loading (diffuse and point

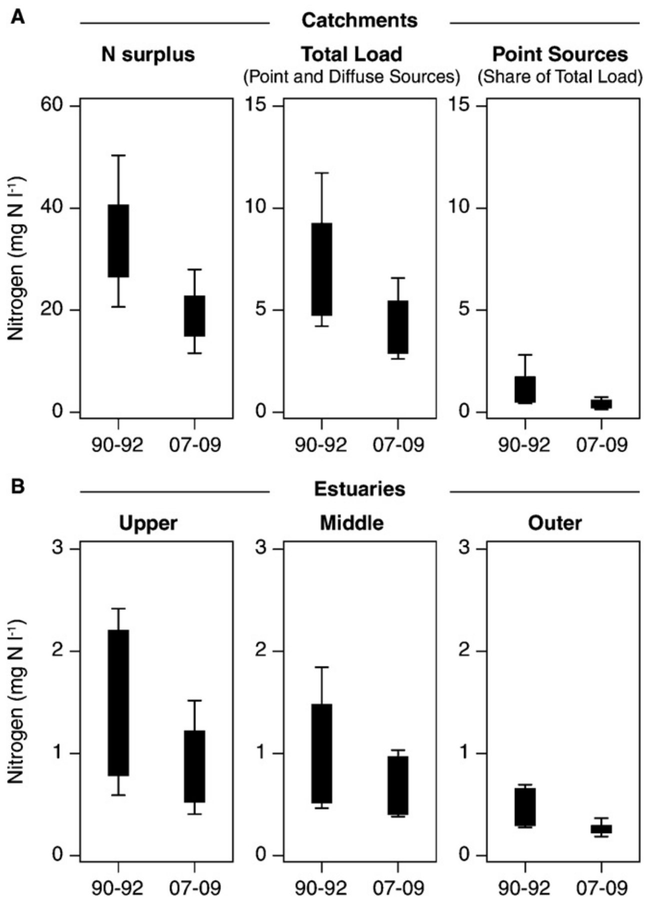

Fig. 3 – Box-plot distributions of nitrogen indicators shown as discharge-weighted concentrations (N divided by the freshwater runoff) for N surplus, total N load, including both point sources and diffuse sources (A) nitrogen concentrations in different parts of the estuaries (B) during the period 1990–92 to 2007–09. Boxes mark the inter-quartile range and whiskers show 10% and 90% percentile values.

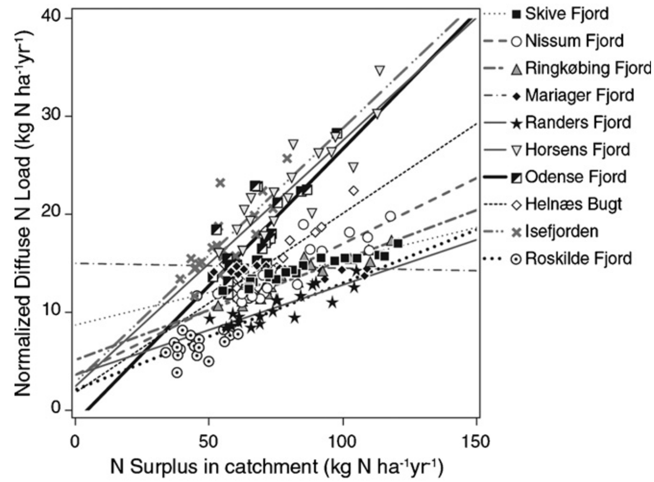

Fig. 4 – Relationships between nitrogen surplus and normalised nitrogen load from diffuse sources for the 10 study catchments during the period 1990–2009.

sources) to the 10 estuaries was recorded (mean: 39%), but with marked variations between the individual catchments causing very different responses in the N concentrations in inlet fresh water to the estuaries (18–55%; supplementary material C).

Part of the reduction in N load can be explained by a diminished annual discharge of N from catchment point sources, amounting to 3500 t N in total and thus constituting a significant reduction of point source related contributions to the N concentrations in freshwater inlets to the estuaries (Fig. 3A). However, the major part of the reduction in N loads (Fig. 3A) can be ascribed to lower loads from diffuse sources within the catchments. During the period 1990–2009, the total diffuse load to the 10 estuaries was reduced by an average of 34%. In the estuaries, the reduced N loads were manifested as significantly lower ($P < 0.001$) N concentrations (24–62%; supplementary material C). In sum, both stream and upper estuary concentrations nearly halved over the time frame investigated. Our results are in line with the findings of Bouraoui and Grizzetti (2011) who in their analysis of trends in nitrate-N loading of European rivers detected a declining trend in 30% of the analysed rivers, the most pronounced reductions occurring for rivers draining to the north sea during the period 1991–2004. Also, our findings are corroborated by those of Kronvang et al. (2008) for smaller Danish catchments.

4.2. *Relationships between the nitrogen surplus and diffuse load to estuaries*

Normalised diffuse N load was related to N surplus in the 10 investigated catchments during the period 1990–2009 (Fig. 4). Catchment-specific relationships were identified, showing that the catchments responded individually to the NAPs (supplementary material D). Notable quantitative responses were observed for the Horsens Fjord, Isefjord and Odense Fjord catchments where the N surplus (field) was reduced by 30–52 kg N ha^{-1} yr^{-1} during the period 1990–2009, yielding normalised N load reductions to the estuaries within the range of 10–14 kg N ha^{-1} yr^{-1}, whereas no response could be traced for the Mariager Fjord catchment (Fig. 4). Rather weak responses were also found in the Randers Fjord and the Roskilde Fjord catchments where N surplus (field) declined by 24–50 kg N ha^{-1} yr^{-1}, yielding a reduction of the normalised diffuse N load of *ca.* 3–5 kg N ha^{-1} yr^{-1} during the period 1990–2009 (Fig. 4).

The differences in catchment responses (slope of relationships shown in Fig. 4 and supplementary material D) are probably related to catchment-specific variations in N removal in groundwater as a consequence of varying geology (Bremen et al., 2002; Hansen et al., 2009), differences in N retention in surface waters as well as time lags in the groundwater flow (Kronvang et al., 2008; Bouraoui and Grizzetti, 2011).

Nine catchments showed significant linear relationships between N surplus and normalised diffuse N loadings despite the fact that time delays in groundwater may be of importance in all catchments. Normally, groundwater time delays imply that water has passed the redox zone and that all nitrate has been removed, but for at least two study catchments this does

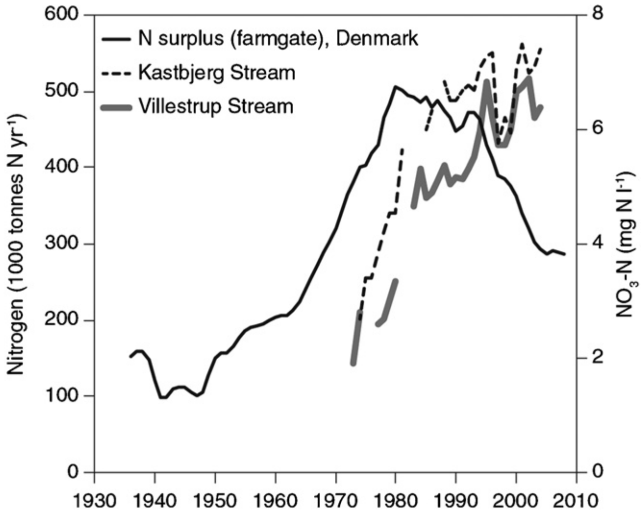

Fig. 5 – Example of historical trends in nitrogen surplus (farm gate balances) and nitrate-N concentrations in two streams draining to the Mariager Fjord catchment.

not hold true. The Mariager Fjord catchment showed no response to reductions in the N surplus; thus, high normalised diffuse stream loads occurred throughout the study period (Fig. 4). The same phenomenon was partly observed in the Skive Fjord catchment being located in the same region. All the other catchments demonstrated a y intersection value at low N load values, signifying an estimated mean N concentration of 1.24 ± 0.46 mg N L^{-1}, which is near to the mean background N concentration measured in 2010 in nine minor Danish streams that all drain catchments with minimum anthropogenic disturbance (1.27 ± 0.55 mg N L^{-1}). The reason for the high y intersection value for the Mariager Fjord catchment is likely a combination of time lags and dominance of old and oxic groundwater. Hansen et al. (2011) have demonstrated a low trend of nitrate concentrations in old and oxic groundwater in Denmark during the last 20 years. Moreover, Basu et al. (2010) proposed that water quality problems in receiving surface waters will persist until the legacy stores are substantially depleted. Only two of our 10 analysed paired catchments and estuaries seem to have a legacy store of nitrate in the groundwater aquifers, demonstrating that in most cases surface waters respond directly (<5 years), but to a varying degree, to the implementation of action plans targeted at reducing agricultural N pollution.

For the Mariager Fjord catchment, longer time series on the responses of N surplus and total N concentrations in two monitored streams are shown in Fig. 5. The historical trends in the N surplus show a peak in the early 1980s, whereas the concentration of nitrate-N in the Villestrup and Kastbjerg streams shows a delayed peak in the late 1990s (Fig. 5). The N responses to policy measures in the Mariager catchment thus exhibit a decadal time delay, possibly explainable by the dominance of chalk aquifers discharging old groundwater to surface waters as shown by a CFC-dating of groundwater revealing a mean age of 25 years according to Wiggers et al. (2000). Moreover, the groundwater is oxic, thus preventing denitrification of nitrate N.

4.3. *Relationships between catchment nitrogen load and nitrogen concentrations in estuaries*

The observed total N concentrations in the 10 estuaries were significantly linked to the observed total N loads expressed as flow-weighted total N concentrations in inlet freshwater to the estuaries for all catchments (Table 4). These results are consistent with those reported in Carstensen and Henriksen (2009). The slopes in the relationships indicate that the estuaries are affected quite differently by nutrient inputs from land due to variations in hydrologic residence times across estuaries, ranging from a few days in Randers Fjord to about 3 months in Ringkøbing Fjord (Rasmussen and Josefson, 2002), in the location of monitoring stations along the salinity gradient, and in total N concentrations at the mouth of the estuary. Estuaries with short retention times will typically display a linear mixing pattern of nutrients along the salinity gradient, whereas estuaries with longer retention times typically have a curvilinear relationship with salinity because

Table 4 – Annual total nitrogen concentration ($N_{estuary}$ mg N L^{-1}) in the upper or middle reaches of 10 estuaries as a function of the catchment total nitrogen load (N_{inlet}) expressed as discharge-weighted total nitrogen concentration (mg N L^{-1}) in inlet freshwater from diffuse and point sources. $N_{estuary} = a \times N_{inlet} + c$. C.L.: 95% confidence limits for the parameter estimate.

Estuary 1990–2009	a	95% C.L.	c	95% C.L.	r^2
Skive	0.469***	0.313; 0.625	−1.2209	−1.977; −0.463	0.69
Nissum	0.562***	0.408; 0.716	0.122	−0.446; 0.690	0.78
Ringkøbing	0.713***	0.618; 0.807	−1.130	−1.444; −0.816	0.93
Mariager	0.292**	0.097; 0.487	−0.714	−2.061; 0.634	0.35
Randers	0.605***	0.484; 0.725	−0.423	−0.860; 0.0153	0.87
Horsens	0.062***	0.043; 0.082	0.0795	−0.081; 0.240	0.73
Odense	0.191***	0.127; 0.255	0.113	−0.376; 0.601	0.69
Helnæs Bugt	0.095***	0.073; 0.118	−0.0177	−0.156; 0.121	0.84
Isefjord	0.027**	0.011; 0.043	0.273	0.120; 0.425	0.41
Roskilde	0.086**	0.040; 0.131	0.533	0.269; 0.797	0.46

** $P < 0.01$.
*** $P < 0.001$.

Table 5 – Catchment-specific reductions of the nitrogen surplus and the diffuse nitrogen load, and the cost per kg N of reducing the N load from diffuse sources.

Estuary 1990–2009	Nitrogen surplus		Diffuse nitrogen load	
	Reduction of nitrogen surplus 10^3 t N	Costs 10^6 Euro	Reduction of diffuse N load 10^3 t N	Cost/kg N reduced load Euro/kg N
1. Skive	15.7	204	1.077	189
2. Nissum	8.7	113	1.187	96
3. Ringkøbing	19.0	247	1.993	124
4. Mariager	2.8	37	–	–
5. Randers	16.2	210	1.532	137
6. Horsens	2.7	35	0.730	48
7. Odense	3.5	46	1.162	40
8. Helnæs	0.67	8.7	0.137	64
9. Isefjord	2.3	30	0.729	41
10. Roskilde	2.8	37	0.359	104

N removal increases with retention time through denitrification and permanent burial. Thus, policy measures adopted to reduce N surpluses in the catchments will, as shown for 8 of our analysed catchments with no time lags in groundwater, have a direct (<5 years) effect on the N concentrations in estuarine waters, and this is the main cause of the declining (i.e. improving) nutrient trends observed in Danish estuarine and coastal waters (Carstensen et al., 2006). Slow responses of the natural environment to changes in N loads from land following adoption of mitigation measures in catchments are a major factor to be considered when predicting responses (Maier et al., 2009). Moreover, the responses observed in estuaries are modulated by residence times such that longer residence times stimulate the retention of N within the estuary and thus reduce the export of N to the open coastal waters.

4.4. *Costs of NAPs and future directions*

Calculations of the costs per kg reduction in the N load to estuaries arising from the implementation of NAPs during the past 25 years are displayed in Table 5 and clearly show substantially higher expenses in catchments with high than in catchments with low N attenuation. Introduction of general mitigation measures to reduce diffuse N losses from Danish agriculture is indeed costly; however, at the same time farmers have benefited economically from the reduced use of chemical fertilisers due to the efficient use of N in manure.

Even though N loadings and N concentrations have been reduced considerably to most of the studied estuaries, the target N concentration for obtaining good ecological quality under the EU WFD has not been met, not even in the catchment and estuaries showing the most pronounced responses to agricultural N surplus reductions (Environment Centre Odense, 2007; Hinsby et al., 2012). As an example, model estimations show that future N loadings have to be reduced by additionally 48% and 55% from the 2001–2005 figures in order to obtain good ecological quality in the estuaries of respectively, Odense and Horsens (Environment Centre Odense, 2007; Hinsby et al., 2012). In the newly adopted river basin management plans under the EU WFD, targeted regional mitigation measures to reduce N surplus and N loads have been implemented in agricultural areas downstream of larger lakes (as in the Randers and Roskilde Fjord catchments) to achieve the strongest possible effects of reduced N surplus on the N load. Similarly, we foresee that in the 2nd and 3rd river basin management plans to be adopted under the EU WFD, also local general (N surplus reduction) and targeted (wetlands, buffer strips, controlled drainage) mitigation measures will be implemented following an analysis of the N removal potential in groundwater and surface waters (Kronvang et al., 2008; Stutter et al., 2012).

5. Conclusions

The Danish regulatory approach to diminish agricultural N pollution during the last 25 years has mainly involved implementation of general mitigation measures. On a national scale, the outcome has been successful—the total land-based N load to Danish coastal waters has been reduced by approximately 50% coming from an 18,500 t N reduction from point sources and a 32,000 t N yr^{-1} reduction from diffuse sources during the period 1990–2009. The regional responses have, however, varied widely. To elucidate this, we conducted an analysis of agro-environmental indicators from 10 catchments and their estuaries representing regional differences in climate, soil type and geology.

We have documented new evidence of N attenuation in catchments and their estuaries (N-removal and time lag) by demonstrating significant and highly variable regional reductions in N during the last 25 years in response to the adopted policy measures. We have found examples of 6 catchments responding directly (<5 years) with marked and significant reductions in the N load to the estuaries to the adopted agricultural mitigation measures and of two catchments showing a direct but less marked response due to the presence of large lakes. Finally, two other catchments show almost no response in N load due to lag times in flows of nitrate-rich groundwaters.

Our catchment and estuary analysis documented a difference factor of 3–4 in the costs of achieving N load reductions from diffuse sources between the 10 analysed catchments. We therefore conclude that attenuation of N (removal and time lags) in catchments may significantly impact the outcome of management plans targeted at obtaining N load reductions to estuaries and/or may delay the responses so that trend reversal is not detected for decades.

Further reductions of present N loads to Danish estuaries during the implementation of the WFD are required. To assist in reducing the costs of combating land-based diffuse N loads, we recommend the development of cheap methods providing improved insight into variations between catchments and also into the spatial variation in N retention within catchments. The results of our investigation show that introduction of targeted mitigation measures in regions or sub-catchments with low N removal is the most cost-effective way (cost/kg N reduced) to obtain N reductions in estuaries.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.envsci.2012.08.009.

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